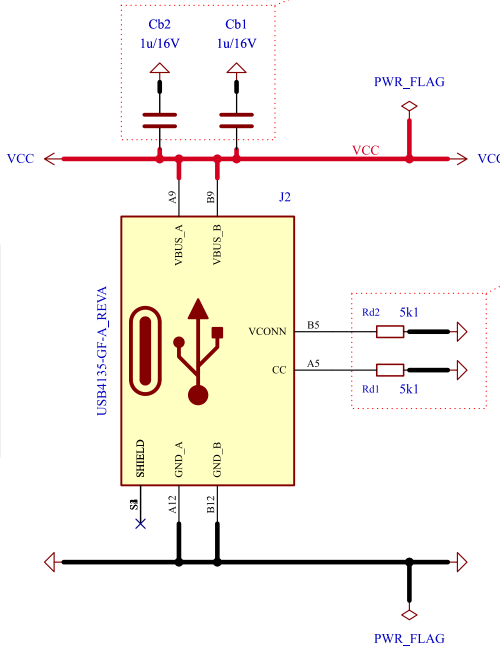
[4] USB connector



DESIGN NOTE:
Cb1, Cb2
bulk capacitor

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Table 4-2 VBUS Source Characteristics

	Minimum	Maximum	Notes
VBUS Leakage Impedance	72.4 kΩ		Leakage between VBUS pins and GND pins on receptacle when VBUS is not being sourced.
VBUS Capacitance		3000 µF	Capacitance for source-only ports between VBUS and GND pins on receptacle when VBUS is not being sourced.
		10 µF	Capacitance for DRP ports between VBUS and GND pins on receptacle when VBUS is not being sourced.

DESIGN NOTE:
Per USB-IF specification, a sink role is established by placing 5.1 kΩ pull-down (Rd) resistors on CC1 and CC2.

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Table 4-25 Sink CC Termination (Rd) Requirements

Rd Implementation	Nominal value	Can detect power capability?	Max voltage on pin
± 20% voltage clamp[1]	1.1 V	No	1.32 V
± 20% resistor to GND	5.1 kΩ	No	2.18 V
± 10% resistor to GND	5.1 kΩ	Yes	2.04 V

Note:
1. The clamp implementation inhibits USB PD communication although the system can start with the clamp and transition to the resistor once it is able to do USB PD.

www.usb.org/sites/default/files/USB%20Type-C%20Spec%20R2.0%20-%20August%202019.pdf

Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 5ba43f3
	Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
Sheet Title: USB connector	File Name: Section A - USB connector.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Section A - USB connector/		Reviewer:	Size: A4	Sheet: 4 of 9